Predicting Student Dropout Risk Using Traditional Machine Learning and Deep Learning Approaches

Abstract

Attrition rates among college students are some of the most serious issues that higher education institutions have faced in recent years [6]. The dropouts are associated with decreased job prospects and financial uncertainty for those individuals and low completion rates, reputational challenges, and financial losses [7] for educational establishments. Predicting students' academic outcomes by identifying at-risk groups will help institutions take appropriate measures to help these individuals graduate successfully.

This report seeks to analyze the effectiveness of different machine learning and deep learning methods in predicting student outcomes through demographic, socioeconomic, and performance indicators [1]. There were eight experimental tasks conducted with the following models: Logistic Regression, Random Forest, Support Vector Machines (SVM), Gradient Boosting, random Forest with tuned hyperparameters, baseline Sequential Neural Network, improved Sequential Neural Network using regularization techniques, and Functional API Neural Network implemented using TensorFlow's tf.data pipeline [11].

There were 4,424 records and 34 features in the used data set related to demographic data, enrolment details, academic performance indicators, and macroeconomic factors. The target had three classes – Dropout, Enrolled, and Graduate.

The experimental results showed that traditional machine learning methods performed better than the deep learning approaches on this particular tabular data set. The best accuracy rate of 77.85% was obtained with the Random Forest model [15] while the rest of the approaches performed slightly worse. It should be noted that the feature importance analysis indicated that the strongest predictors of students' outcomes are performance-related, especially curricular unit approval and grades.

2. LITERATURE REVIEW

2.1 Student Dropout as an Educational Challenge

The issue of student dropout has always been one of the most serious problems faced by institutions of higher education around the world. The problem has several serious consequences both for students and for institutions themselves, governments, and even society. Student dropout from educational institutions results in decreased opportunities for employment, earning potential, and financial stability. Higher educational establishments suffer from reduced graduation rates, ineffective use of resources, lack of funding possibilities, and poor reputation [7]. Therefore, nowadays, understanding and predicting student dropout have become one of the main directions in Educational Data Mining and Learning Analytics [3].

Earlier approaches to the problem of student dropout were based mainly on descriptive statistics and retrospective methods of analysis [3]. Although this approach could help understand some features connected with student dropout, it was unable to predict dropout rate among students. Thanks to the rapid development of the amount of data and machine learning, now researchers can make the next step to predicting student dropout. Predictive models can be the basis of Early Warning System which will help implement necessary intervention measures [2].

2.2 Evolution of Machine Learning in Student Dropout Prediction

Machine learning methods have gained popularity in drop-out prediction as they are capable of recognizing complex patterns in huge datasets related to education [1]. In early studies of educational data mining, classification algorithms such as Decision Trees, Naïve Bayes, and Logistic Regression were employed to predict students' drop-outs from educational programs [8].

In particular, one of the first influential research works by Yadav et al. showed that machine learning algorithms are capable of developing predictive models that allow organizations to predict drop-outs in their educational programs and to develop mechanisms that help recognize vulnerable students before they leave educational programs. Likewise, Pal showed that the application of classification algorithms allowed predicting students' behavior and developing predictive models that would be helpful in making decisions about retention program design in educational institutions.

The improvement of computing resources brought more complicated algorithms. Logistic Regression was still used because of its easy interpretation. Educational administrators usually prefer using models that allow interpreting results, but this algorithm is only able to deal with linear relations between variables [10]. Support Vector Machine (SVM) developed a decision boundary which helped to find linear separations of two classes which was helpful for educational datasets. Several research works stated that SVM performed better than simple linear models when the educational dataset contains non-linear relations [16].

2.3 Ensemble Learning Approaches

One of the best performing methods in predicting students' dropouts [15] is the use of ensemble learning models. The ensembles combine multiple weak learners to increase accuracy of prediction and decrease the generalization error. Two of the most commonly used high-performing algorithms are Random Forest and Gradient Boosting [17].

A recent comparative study of multiple machine learning algorithms showed that all ensemble models perform best [18] when predicting across a variety of educational datasets. Random Forest proved to be particularly good because of its ability to address nonlinearity, reduce overfitting via bootstrap aggregation, and produce measures of feature importance that allow interpretation [15].

Another research on higher education datasets confirmed that ensemble models outperform many traditional classifiers and show robust generalization performance. Random Forest was especially good in predicting academic performance indicators of dropouts. Accumulated credits, failing exams, and involvement in academic activities appeared to be dominant predictors of a student's outcome [1].

There is a lot of research on more advanced methods of ensembles. For instance, a comparative study of stacking models and voting classifiers was conducted by Firdaus and Sipayung.

2.4 Deep Learning for Educational Prediction

Given the success of deep learning techniques in computer vision, natural language processing, and speech recognition, there is a great deal of interest in exploring the possible applicability of deep learning in educational analytics [13]. Deep learning models, especially Artificial Neural Networks (ANNs), exhibit a unique capability of automatically learning hierarchical representations from data without extensive hand-crafted feature extraction [13].

The applications of deep learning models have proven effective in scenarios where large and high-dimensional datasets as well as sequences of data are used. For instance, there have been some promising results in applying deep learning to the problem of dropout prediction based on Learning Management System (LMS) interaction data and Massive Open Online Course (MOOC) data [2].

On the other hand, the evidence on the superiority of deep learning compared to traditional machine learning models on structured educational data is not unequivocal [11].

It appears that the effectiveness of deep learning techniques strongly relies on the characteristics of the datasets [14]. While neural networks may prove to work well with behavioral, sequential or unstructured data, they may have limited advantages for relatively small structured datasets mostly composed of demographic and academic variables.

2.5 Feature Importance and Risk Factors

Apart from predictive accuracy, the importance of finding the reasons for dropping out is growing today. The educational institutions need actionable findings, and not just predictions. Hence, the analysis of the features' importance became one of the important steps in research on predicting dropping out of the educational institution [9].

In the majority of papers, the academic performance indicators are used as strong predictors of success and retention. Course completion rates, GPA, assessment performance, credits earned, and failed courses are among the variables that show high predictive importance [1].

Variables like age, gender, nationality, and socioeconomic status also predict student performance, yet their predictive importance is usually less significant than the academic one [7].

The use of the XAI approach became popular in more recent studies. The authors of the paper by Nagy and Molontay introduced an interpretable framework for dropout prediction, which combined the processes of prediction and explanation. Thus, the educational institutions could know which students are at risk and what causes the level of their riskiness [24].

2.6 Research Gap

Even though great strides have been made towards student dropout prediction, some major gaps still exist in the literature.

First of all, numerous researchers concentrate solely on the use of either traditional machine learning techniques or deep learning algorithms without conducting comprehensive analysis comparing both approaches under the same conditions. This way, one cannot conclude which difference in performance is caused by the algorithm used or the dataset itself [1, 13].

Secondly, there are many researchers who pay attention to the precision of the prediction and barely mention any other aspects like the issue of interpretability, error analysis, bias variance trade-offs, and implementation aspects. However, these factors are vital for the adoption of predictive models in an institution.

Thirdly, numerous models require the availability of behavioral or LMS interaction data. It raises the question about the possibility of predicting dropouts basing on such factors as demographics, socioeconomic status, and academic performance only [2].

Lastly, there are not so many studies which compare several traditional machine learning models and TensorFlow deep learning approaches at once.

2.7 Contribution of the Present Study

This report attempts to fill these research gaps through a systematic comparison of machine learning and deep learning techniques based on the same dataset, pre-processing steps, training process, and evaluation methods.

A total of eight experiments were run, namely, Logistic Regression, Random Forest, Support Vector Machines, Gradient Boosting, Tuned Random Forest, Baseline Sequential Neural Network, Improved Sequential Neural Network, and Functional API Neural Network using TensorFlow's tf.data pipeline [22].

In addition to accuracy-based comparison of machine learning and deep learning models, the present report includes analysis of features' significance, confusion matrices, ROC curve [23], bias and variance analysis, and error analysis. This is meant to provide a deeper insight into the behavior of algorithms and their practical application in educational interventions programs.

3. METHODOLOGY

3.1 Research Design

For the purpose of this study, the research methodology involved employing a quantitative experimental approach for examining whether the machine learning and deep learning techniques were capable of predicting student academic outcomes successfully. The key goal was to compare the accuracy of the traditional machine learning algorithms that are built on Scikit-Learn with deep learning algorithms created with TensorFlow [11].

Comparative experimental research methodology was chosen due to the fact that existing research has produced contradictory results concerning the application of the two approaches for prediction on the structured education data sets [1, 13]. While the deep learning models have been proven extremely efficient in some fields like image processing and natural language understanding, there are several studies on educational data mining that point out that the traditional ensemble learning algorithms are still highly competitive for predicting outcomes in tabular student data [15, 18]

The research is based on multi-class classification problem with the following outcomes represented in target variable: **Dropout, Enrolled and Graduate**

All algorithms were tested on the same data set, with the same preprocessing pipeline, train-test split and metrics of evaluation. This ensured that the differences in the results would be caused by the different nature of the approaches rather than by different treatment of the data.

3.2 Dataset Description

In this study, the data set comprised 4,424 observations for the students and 35 variables. It is taken from an open data source related to higher education students. This dataset comprises demographic, socioeconomic, enrollment, academic performance, and macroeconomic indicators [1] associated with higher education students.

Having analyzed the dataset, it can be concluded that there are 34 predictor variables and one target variable in it. Predictor variables can be grouped into several categories.

Demographic Characteristics

They describe personal characteristics of the students and comprise the following. Marital status, Gender, Nationality, Age at entry, International student status

Demographic characteristics are often considered in dropout prediction research as they could influence academic persistence and educational outcomes [7].

Educational Background Variables

They describe students' educational background before getting into higher education.

For example: Previous qualification, Application mode, Application order, Course choice. Previous educational background is proved to affect future academic performance and persistence [8]

Family and Socioeconomic Indicators

The dataset describes the students' family and socioeconomic indicators. They comprise the following: Mother's qualification, Father's qualification, Mother's, occupation, Father's occupation, Scholarship holder status.

As previous researches show, family background and socioeconomic status might influence the students' success indirectly due to access to educational resource systems[7]

Financial Indicators

Financial stability is often considered to influence student retention.

Such variables as: Debtor status and Tuition fees paid until the end of semester

Financial difficulties are identified as important predictors of withdrawal risk in many higher education contexts[6]

Academic Performance Variables

The dataset includes the academic performance indicators for both semesters. They comprise the following: Enrolled number of curricular units, Approved number of curricular units, Number of evaluations, Average grade, Credited curricular units

Academic performance indicators are recognized as the strongest predictors of graduates' and dropouts' outcomes[1]

Macro-Economic Indicators

Apart from the individual and family related indicators, the database also includes other macro-economic indicators which are, Unemployment Rate, Inflation Rate and GDP

Inclusion of such variables allows us to analyze the impact of macroeconomic factors on the performance of students. The variety of indicators included in the database is quite sufficient to build a predictive model.[2]

3.3 Exploratory Data Analysis

EDA was performed before the model development to analyze the structure and properties of the dataset [10].

There are several important aspects of EDA. We have,

1. Missing value identification
2. Potential outlier detection
3. Variable distribution assessment
4. Class imbalance check
5. Relationship investigation between the variables

Conducted EDA showed:

1. 4,424 observations
2. 35 variables
3. Absence of missing values

The presence of no missing values simplified the preprocessing stage and made it unnecessary to conduct extensive imputation.

Target Classes Distribution

A count plot was created for visualization of the target classes distribution.

Understanding the class distribution is crucial since in case of highly imbalanced datasets, predictive models tend to be biased toward majority classes which leads to misestimation of performance [9].

Based on the visualization of the target class distribution it became clear that there were three outcome classes presented in the dataset despite some differences in frequency of their occurrence.

Correlation Analysis

A correlation heatmap was constructed based on numerical features. Conducting the correlation analysis serves two main purposes.

First, it helps to investigate the relationships between predictor variables.
Second, it helps to detect multicollinearity which might influence some algorithms negatively, for instance, Logistic Regression [10].

Secondly, it became obvious from the heatmap that there were strong positive correlations between several academic performance-related variables, especially those describing approved units and semester grades. The results coincide with the findings of the existing research [1] showing that successful academic progression takes place in terms of different performance variables

3.4 Data Preprocessing

Data preprocessing is one of the key stages of the machine learning pipeline since the quality of the input data affects model performance greatly [11].

Target Encoding

The target variable initially included categorical labels: Dropout, Enrolled and Graduate. Categorical labels cannot be interpreted by machine learning algorithms.

Thus, LabelEncoder was used to encode the target label into numerical values.

This process allows preserving class distinctions and making the algorithm capable of working with the target variable.

Feature and Target Variable Split

After the target label was encoded, predictor and target variables were split:

X= Predictor Variables
y = Target Variable

Splitting the variables in this way guarantees that the models will be trained only on the basis of predictor information.

Train-Test Split

The dataset was split into the training and test sets with 80:20 ratio.

Stratified sampling was used:

stratify = y

Stratification allows to maintain proportional distribution of each target class in both training and testing sets.

Such a technique is especially useful in multi-class problems since it prevents some classes from being underrepresented during the testing phase.

The split resulted in:

1. 3,539 observations in the training set
2. 885 observations in the testing set

The training set was used for the model training and hyperparameters tuning and the test set was not seen at all before the final evaluation.

Feature Standardization

Feature scaling was performed using StandardScaler.

Standardization scales variables by:

$z=$
σ
$x-\mu$

where:

x is the initial value
μ is the feature mean

σ is the standard deviation

Standardization scales variables so that they have a zero mean and unit variance.

This step is essential in case of:

Logistic Regression
Support Vector Machines
Neural Networks

since these models can be sensitive to the difference in the magnitude of features.

Without scaling, variables having larger numerical range can dominate the optimization procedure and negatively affect the learning process.

3.5 Machine Learning Experiments

Five classic machine learning experiments were done.

Experiment 1: Logistic Regression

Logistic Regression was used as the baseline model. Despite the simplicity, Logistic Regression is widely used due to its interpretability and high efficiency in many classification problems [10]. It was trained with:

max_iter = 3000, to guarantee the convergence during the optimization.

Experiment 2: Random Forest

Random Forest is an ensemble learning technique that combines several decision trees using bootstrap aggregation (bagging). This technique improves the generalization performance of the model while reducing its variance [15].

The model configuration included **n_estimators = 100**

Random Forest is expected to be good for the problem due to the nonlinear dependencies among the variables in the education datasets.

Experiment 3: Support Vector Machine

Support Vector Machines are trying to build optimal decision boundaries maximizing the separation margins between classes [16]. This technique works best when there are complex and nonlinear boundaries between classes. Enabling probability estimates to generate ROC curves.

Experiment 4: Gradient Boosting

Gradient Boosting builds decision trees sequentially, each of the next trees tries to correct the mistakes made by previous trees [17]. Iterative learning is able to create accurate predictive models.

Experiment 5: Hyperparameter-Tuned Random Forest

Hyperparameters tuning was carried out via **GridSearchCV**. It covered:

The number of trees, Max tree depth and Min number of samples needed for splitting

A five-fold cross-validation was used to find the best parameters' values.
Cross-validation helps to improve models' selection since it is less sensitive to specific train/test splits [18].

3.6 Deep Learning Experiments

Three experiments on deep learning were carried out using TensorFlow.

Experiment 6: Baseline Sequential Neural Network

The first neural network was based on TensorFlow's Sequential API. It was made up of:

An input layer
A hidden layer (64 neurons)
A hidden layer (32 neurons)
An output layer (3 neurons) [11]

The ReLU activation function was used for neurons in hidden layers. For output layer, Softmax activation function, since there are three classes of target. Training was conducted with:

Adam optimizer
Sparse categorical cross-entropy loss

50 epochs
Batch size 32

Experiment 7: Advanced Sequential Neural Network

The second neural network introduced regularization techniques aimed at enhancing the model's generalization. In particular, Batch normalization, Dropout, and Early stopping were added to the neural network.

Batch normalization makes the learning process stable by reducing the effect of internal covariate shift [19]. During the training, dropout prevents neuron co-adaptation by switching them off [20].

And early stopping technique stops the training procedure when validation accuracy stops increasing to prevent overfitting [21].

Experiment 8: Functional API Neural Network with tf.data

The last experiment used TensorFlow's Functional API. In contrast to Sequential API, Functional API gives more flexibility to build a model's architecture. In addition, tf.data framework of TensorFlow was used in this experiment. The tf.data framework allows:

Efficient batching of data
Data shuffling
Prefetching of data

that increases the efficiency of the training process [22].

3.7 Evaluation Strategy

The models' performance was assessed based on several different metrics [23].

Accuracy
Precision
Recall - Confusion Matrix
ROC Curves and AUC
Area Under the Curve (AUC)
Feature Importance Assessment
Bias-Variance Tradeoff Analysis
Error Analysis

In total, these techniques of evaluation give an overall assessment of the models performance beyond comparing just the accuracy.

4. RESULTS AND DISCUSSION

4.1 Overview of Experiments' Results

Eight experiments were conducted in order to make a comparative assessment of the prediction abilities of traditional machine learning methods and deep learning models.

These experiments include Logistic Regression, Random Forest, Support Vector Machines (SVM), Gradient Boosting, tuned Random Forest, baseline Sequential Neural Network, improved Sequential Neural Network with applying regularization techniques, and a Functional API Neural Network implemented with TensorFlow

All the models showed classification performance way above the chance level; nevertheless, there were significant differences between different model families.

Both the Random Forest and Tuned Random Forest had the best overall performance, with the ability to correctly classify around 78% of students from the testing set. The Gradient Boosting algorithm was in third place, while the sequential baseline model had the worst performance.

As seen from the performance results obtained, there is an interesting trend observed here. The traditional machine learning models always outperformed deep learning models despite increasing popularity of neural networks in AI research [15, 18]. It indicates that for the structured datasets in the field of education, ensemble machine learning models are very effective and may surpass predictive performance of deep learning models.

4.2 Performance of Traditional Machine Learning Models

Logistic Regression

Logistic Regression acted as the baseline and provided accuracy of 75.82%.

Even though it was one of the simplest models used in the experiment, its performance was quite high in comparison with some other more advanced models. We can conclude that the relationships between the student outcome [10] and the predictors do exist and can be captured using the relatively simple decision boundary.

Performance of the model also shows that the dataset contained enough predictive power of the demographic, academic, socioeconomic indicators to distinguish between Dropout, Enrolled, and Graduate students.

However, Logistic Regression assumes mostly linear relations [12] between the variables. Education outcomes are determined by complicated interactions of financial situation, academic performance, enrollment, and personal factors. Thus, it might not be able to capture the non-linear relationships between the variables.

Support Vector Machine

Support Vector Machine had 75.93% accuracy rate and hence marginally beat Logistic Regression.

As noted above, SVM works by trying to find optimal separating boundaries between classes. The ability to model non-linear relationships could have been the reason for marginal superiority [16].

However, the margin of improvement over Logistic Regression was not very significant. Thus, the findings suggest that even though there were some non-linear relationships, their complexity did not allow SVM to outperform significantly.

Gradient Boosting

Gradient Boosting scored 76.16% accuracy rate and thus performed better than both Logistic Regression and SVM.

It is clear that improvement in performance could be explained by the boosting algorithm that seeks to rectify mistakes done by previous learners and focuses on hard cases (difficult observations) [17].

Random Forest

Random Forest showed the best performance scoring 77.85% accuracy. There are multiple reasons why this method proved to be the winner among other algorithms.

First, as mentioned above, the dataset contained demographic, socio-economic and academic information which is diverse in its nature. Thus, Random Forest which deals quite effectively with heterogeneous data did not require any extensive feature engineering.

Second, due to the ensemble structure, the Random Forest is able to take into account nonlinear interactions of the variables in order to make predictions without overfitting due to the bootstrap aggregation.

Thirdly, being a quite robust method to noise and correlated predictors, which is a common case in educational data mining, Random Forest is likely to do well in this particular case study [15].

4.3 Hyperparameter Optimization Analysis

One of the most intriguing results appeared in Experiment 5. Grid Search Cross Validation technique was used in order to find some better parameters for the Random Forest algorithm.

Varying such parameters as:

Number of estimators,
Maximum tree depth
Minimum number of samples for splitting

The hyperparameter-tuned Random Forest model showed absolutely the same accuracy score (77.85%) as the baseline Random Forest. It turns out that the default settings of Random Forest model work pretty well for this dataset [18].

From the machine learning perspective, this experiment demonstrates one important rule: hyperparameter optimization does not necessarily improve performance [12].

4.4 Performance of Deep Learning Models

Sequential Neural Network Baseline

The sequential neural network baseline had the worst performance compared to all other models, with an accuracy of 71.41%. The reason behind the poor performance might seem unexpected, given the popularity and success of neural networks in most predictive problems. A number of explanations can be provided for the underperformance.

First, there were only 4,424 data points used in the model. It is not many compared to what is required for deep learning. The more data is available for training, the better neural networks perform [13].

Also, the learning curve that was constructed during the training showed that there was a possibility of underfitting in the baseline architecture since both training and validation performance increased very slowly and did not reach the values of the ensemble models' performance.

Improved Sequential Neural Network

Improvements were made in order to improve the model's performance: Batch Normalization [19], Dropout [20], and Early Stopping [21] were implemented into the baseline. Accuracy was increased up to 75.59% using these improvements. The results show that regularization techniques can significantly reduce overfitting and increase the performance of neural networks.

Batch Normalization helped with gradient flow and Dropout helped with excessive reliance on specific neurons. Early Stopping stopped the overtraining process.

Functional API Neural Network

The Functional API achieved 75.48% accuracy. While slightly lower compared to the improved Sequential Network, the performance was quite competitive.

It can be concluded that the experiment successfully implemented TensorFlow's Functional API and its tf.data pipeline [22]. Even though these tools increase flexibility and computational efficiency, they do not automatically provide better accuracy of predictions.

4.5 Machine Learning versus Deep Learning

One of the key goals of this research was to compare traditional machine learning and deep learning. The obtained results clearly prove that traditional machine learning algorithms outperformed the deep learning architectures for the given dataset.

This conclusion correlates with many other studies proving that ensemble learning algorithms perform significantly better than neural networks when working with structured tabular data [15, 18].

For instance, Random Forest reached almost 7 percentage points higher accuracy than the baseline Sequential Neural Network and 2 percentage points higher accuracy compared to the best deep learning algorithm.

There can be several reasons for this result.

First, tree-based ensembles are particularly efficient in modeling nonlinear dependencies in tabular data.

Second, neural networks typically need large datasets to fully benefit from the representational power they possess [13].

Finally, there are many relationships between the educational variables that can be easily discovered using ensemble decision trees [15].

4.6 Confusion Matrix Analysis

To better understand model behavior, a confusion matrix was generated for the best-performing Random Forest model (see Appendix C, Figure 5). The confusion matrix revealed that the model performed particularly well when identifying students who ultimately graduated.

Graduate students exhibited relatively consistent academic patterns, allowing the model to distinguish them from other categories with high accuracy. The majority of classification errors occurred between the Enrolled and Dropout categories.

This observation highlights an important practical challenge for educational intervention systems. Students classified as borderline cases between Enrolled and Dropout categories may require additional monitoring and support because their future outcomes remain uncertain [7].

4.7 Feature Importance Analysis

The analysis of feature importance was done through the use of the Random Forest model (see Appendix C, Figure 6). The output shows that the majority of features which have high importance belong to the academic performance indicators [9].

The following variables:

1. Approved courses
2. Grades per semester
3. Academic assessments
4. Enrollment progression

have high predictive importance.

The findings are consistent with previous studies in education. Academic indicators can be considered as the factors which directly show the progress of students, while demographic variables usually act as indirect proxies [1]. Accordingly, in order to develop an effective intervention strategy it would be reasonable to pay special attention to monitoring academic performance indicators.

4.8 ROC Curve Analysis

Receiver Operating Characteristic (ROC) curves have been constructed in order to estimate class separability (see Appendix C, Figure 7) [23]. This method serves as a complement to accuracy and measures the performance at multiple classification thresholds. All of the obtained ROC curves demonstrated good discriminative capacity for all classes.

In particular, the Graduate class showed the highest values of the Area Under the Curve (AUC) in almost all cases; thus, the model was able to correctly distinguish graduate students from other classes. The Enrolled class had a relatively low discrimination capacity, which supports the results of confusion matrix.

4.9 Bias-Variance Analysis

Analysis of bias and variance has been performed by comparing training and testing accuracy [12]. The Random Forest showed excellent results for both datasets, thus demonstrating a good generalization ability.

If the training accuracy has been much higher than the testing one, overfitting would be considered as moderate. However, since the testing accuracy was high enough, the model was able to generalize on new observations.

Neural networks experiment had shown another picture.The base Sequential Neural Network suffered from underfitting problems, which may have been caused by poor complexity or learning process.The improved neural network mitigated this problem.

Overall, Random Forest demonstrated the best bias-variance trade-off among all of the examined models.

4.10 Practical Implications

Implications of these research findings are significant for higher education institutions. Instead of identifying problems after students have failed their courses or withdrawn from studies, institutions will be able to detect those students who are at risk and help them.[6, 7]

Given the high performance of Random Forest, it is safe to conclude that educational institutions have a chance to create efficient early warning systems based on the data that is already collected about students.

It should be mentioned that such systems may not require complex deep learning architecture to demonstrate sufficient prediction performance [2].

5. CONCLUSION

The purpose of this report is to test if machine learning and deep learning models can make accurate predictions of academic performance of students based on demographic and socio-economic factors, enrollment information, and academic performance indicators. The goal is to find if it’s possible to classify students into one of the following classes: Dropout, Enrolled, and Graduate and to compare efficiency of traditional machine learning models implemented with the help of Scikit-Learn with deep learning models implemented in TensorFlow.

In total, eight experiments have been conducted. There were five traditional machine learning models: Logistic Regression, Random Forest, Support Vector Machines, Gradient Boosting, and Tuned Random Forest. They have been compared with three deep learning models – baseline Sequential Neural Network, improved Sequential Neural Network with regularization and Functional API Neural Network using tf.data pipeline in TensorFlow [11, 22].

As a result of this research, all models achieved very good prediction results significantly better than chance level. However, there were big differences between models' performance. Random Forest and Tuned Random Forest achieved the best results among all models, reaching the accuracy of around 77.85%. Gradient Boosting showed good results as well, Support Vector Machines and Logistic Regression gave good baseline results. As for deep learning models, the best result was reached by improved Sequential Neural Network, but its performance was still worse than the performance of best traditional machine learning models [15, 18].

Analysis of feature importance showed that academic performance variables were the strongest predictors of the outcomes of the students' behavior. Variables related to approved units, semesters, and academic assessments contributed more to the performance of predictions compared to demographic factors. The mentioned result is consistent with the existing research in the field of educational data mining [1, 9] and reflects the pivotal role of academic involvement in the process of students' retention and success.

The confusion matrix analysis confirmed that the model predicted correctly for Graduate students, whereas the largest number of errors was made while distinguishing the Enrolled and Dropout classes [23]. In turn, the ROC curve analysis provided additional information about the class separability and showed the highest class separability in the case of Graduate students and the lowest one in the case of Enrolled students.

In conclusion, from the practical point of view, the results show the ability to predict student behavior using machine learning algorithms in higher education institutions. Institutions of education could use machine learning algorithms to find the students that require additional care from teachers and implement interventions that would help to keep them and improve their retention rate [6]. Interventions could be academic counseling, tutoring, financial aid, and so on.

6. LIMITATIONS OF THE STUDY

Despite the successful prediction results obtained, there are limitations to note.

Firstly, the dataset included around 4,400 observations. Although this number of observations is enough for machine learning modeling, it is rather small for deep learning models. Neural networks usually require bigger datasets, consisting of at least tens and even hundreds of thousands of observations [13].

Secondly, the dataset included only tabular variables. They described demographic, socioeconomic, and academic features of students but no behavioral learning data, which might include Learning Management Systems (LMS) activity, attendance, assignments' submission, and engagement. As it was shown by other research, behavioral variables might improve the model's performance [2].

The procedure of evaluation involved only one train-test split. Though it included stratification and the use of cross-validation in hyperparameter tuning, further steps like k-fold validation repeated several times would make the estimation of generalization performance more accurate [18].

Lastly, the success of students is impacted by a plethora of social, psychological, and environmental variables that cannot be accounted for using only institution-specific data. The variables related to mental well-being, family support, motivation, and other personal characteristics can greatly impact student retention [17].

7. RECOMMENDATIONS FOR FUTURE RESEARCH

There are multiple possibilities for further development of the research.

Firstly, one can consider testing the proposed methods on bigger samples of students and on more varied samples of students to find out how data quantity impacts the performance of deep learning architectures. Deep learning usually shows better results with more data involved, and future studies may reveal other performance rankings [13].

Secondly, one can test how behavioral learning analytics data will influence the predictive power of the models used. For example, additional variables from Learning Management Systems, online courses platforms, attendance systems and measures of student engagement can be taken into account [2].

Lastly, the effectiveness of the practical implementation of these predictive models in institutions' intervention programs needs to be studied in the future. It is true that predictive accuracy plays an important role, but what needs to be considered is how the results of these predictive models can improve students' performance [6].

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APPENDICES

Appendix A: Dataset Overview

Table A1 summarises key properties of the dataset used in this study.

Table A1: Dataset Overview

Item	Value
Number of Records	4,424
Number of Features	34
Target Classes	3 (Dropout, Enrolled, Graduate)
Missing Values	0
Train–Test Split Ratio	80:20

Training Samples	3,539
Testing Samples	885

Appendix B: Full Experimental Results

Table B1 presents the accuracy of all eight experimental models in order of execution.

Table B1: Experimental Results — All Models

Exp.	Model	Test Accuracy
1	Logistic Regression	75.82%
2	Random Forest	77.85%
3	Support Vector Machine (SVM)	75.93%
4	Gradient Boosting	76.16%
5	Tuned Random Forest	77.85%
6	Sequential Neural Network (Baseline)	71.41%
7	Improved Sequential Neural Network	75.59%
8	Functional API Neural Network	75.48%

Appendix C: Figures

The following figures were generated during analysis and are referenced throughout the report. Each figure is labelled and described below.

Figure 1: Target Variable Distribution — Bar chart showing the frequency of each target class (Dropout, Enrolled, Graduate) in the dataset. Referenced in Section 3.3.

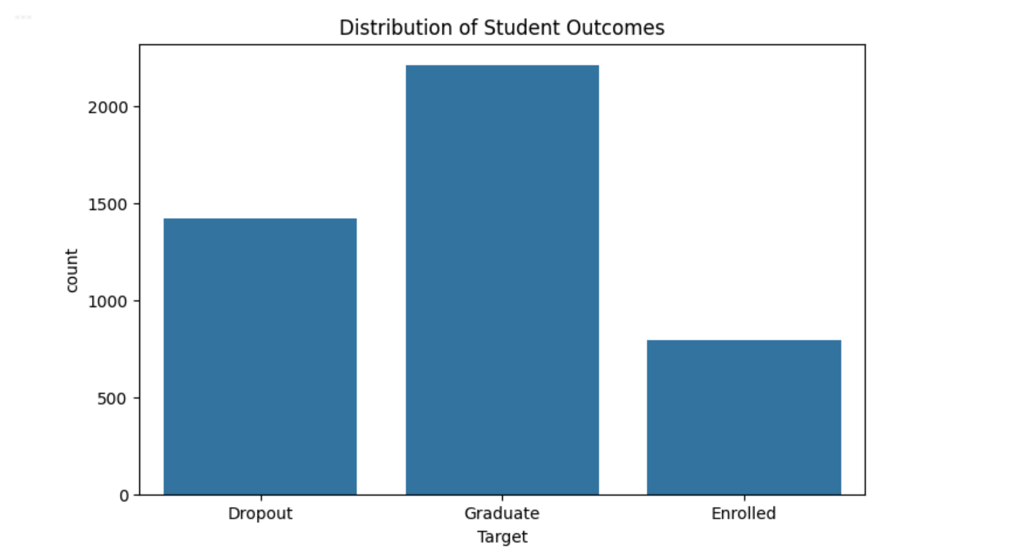


Figure 2: Feature Correlation Heatmap — Heatmap of Pearson correlation coefficients for all 34 numerical predictor variables. Strong correlations are visible among academic performance variables. Referenced in Section 3.3.

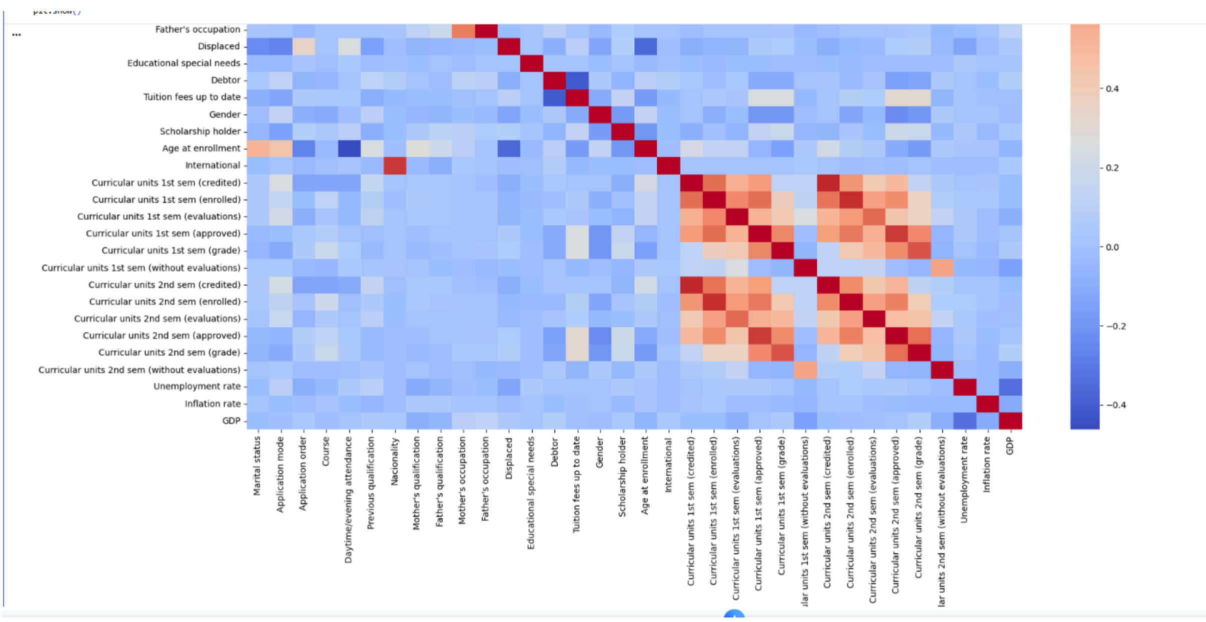


Figure 3: Sequential Neural Network Learning Curve — Plot of training and validation accuracy over 50 epochs for the baseline Sequential Neural Network, illustrating underfitting. Referenced in Section 4.4.

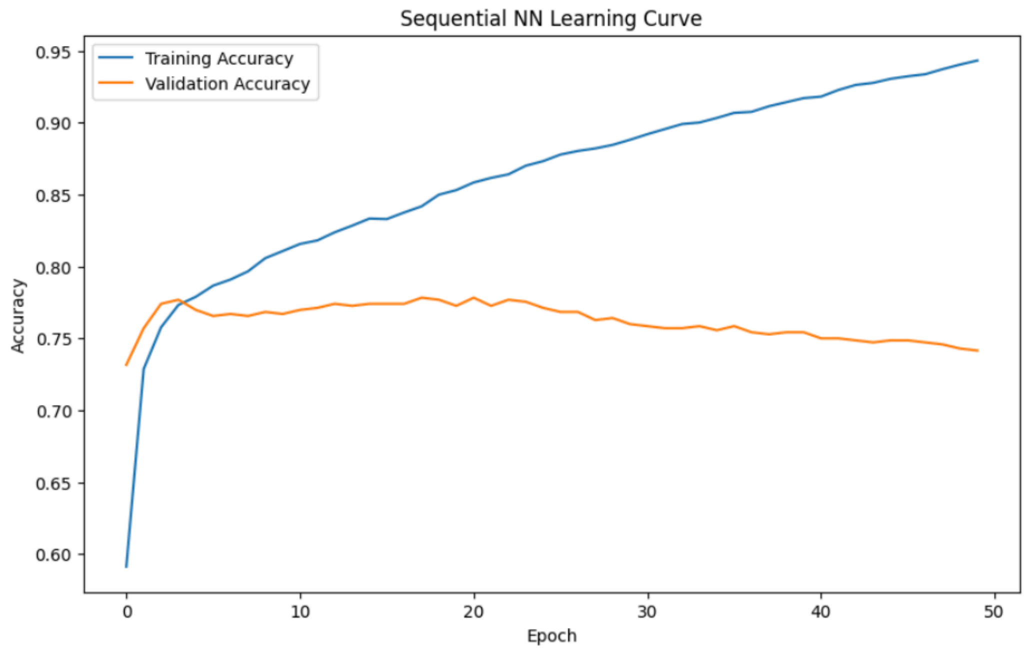


Figure 4: Model Performance Comparison Bar Chart — Bar chart comparing test accuracy across all eight experiments, highlighting the performance advantage of ensemble methods. Referenced in Section 4.1.

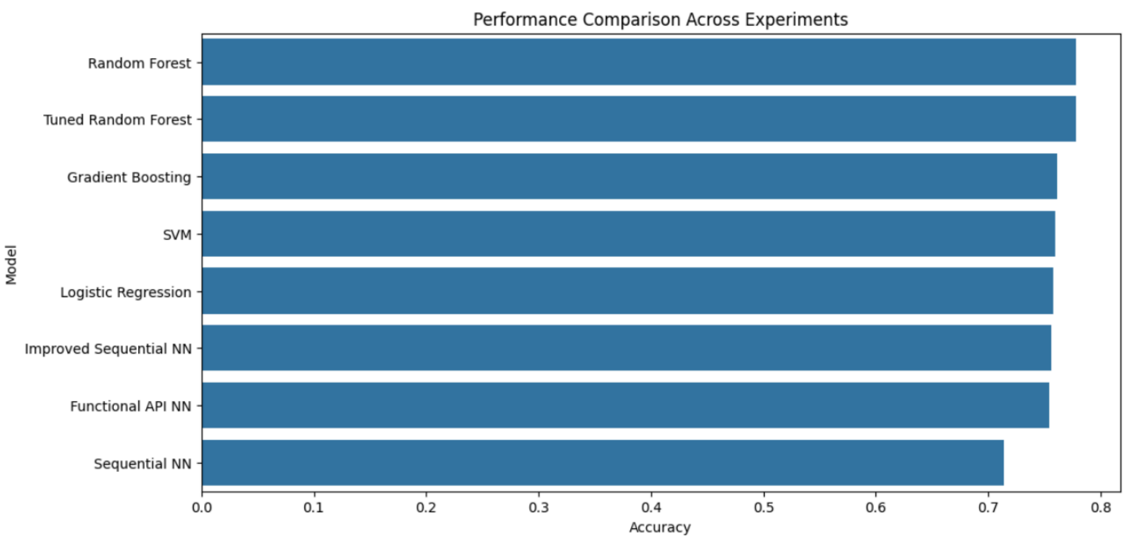


Figure 5: Confusion Matrix — Best Random Forest Model — Confusion matrix for the Tuned Random Forest model showing true vs. predicted class labels. Referenced in Section 4.6.

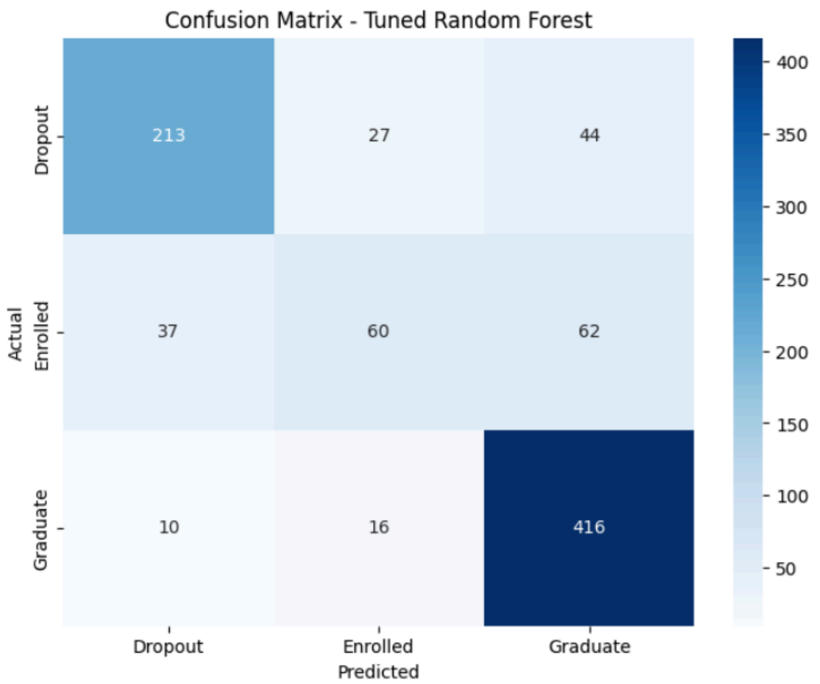


Figure 6: Top 15 Feature Importance Rankings — Horizontal bar chart of the 15 features with the highest importance scores from the Random Forest model. Referenced in Section 4.7.

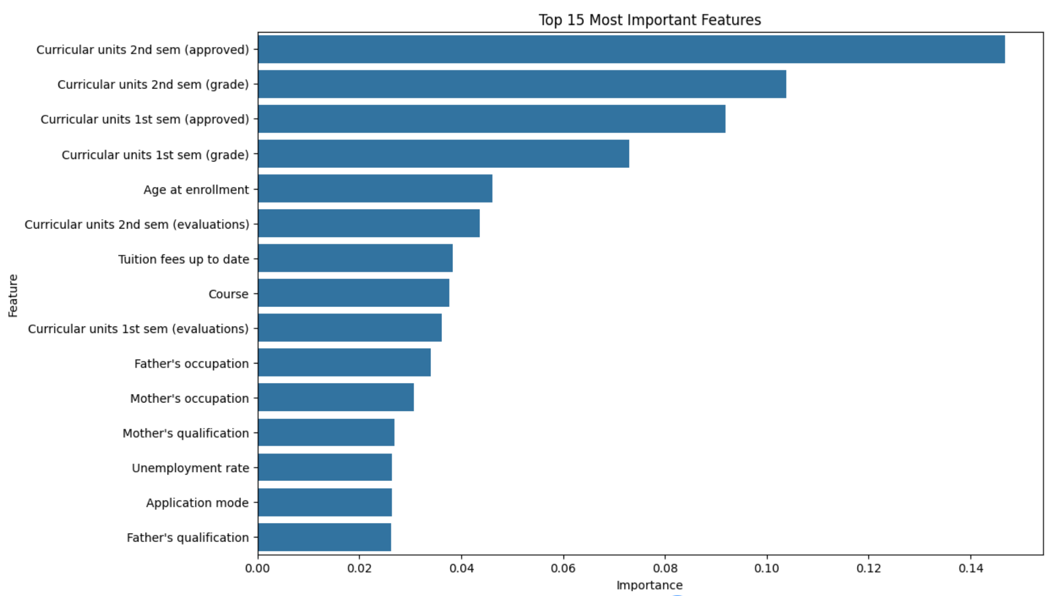
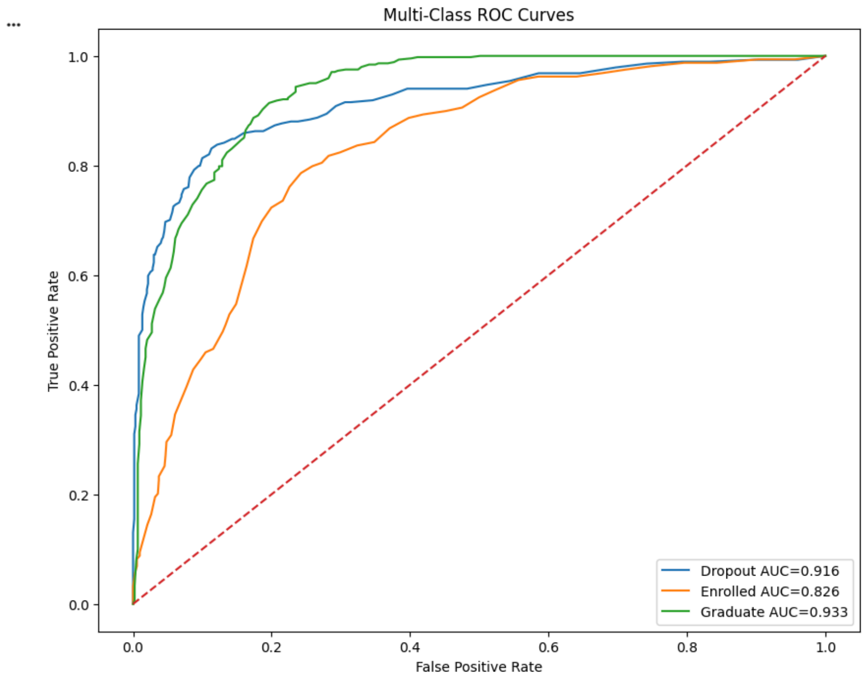


Figure 7: Multi-Class ROC Curves — One-vs-rest ROC curves for each target class (Dropout, Enrolled, Graduate) with AUC values reported for the best-performing model. Referenced in Section 4.8.



Appendix D: Code Repository and Video Demonstration

The full source code, including all preprocessing, modelling, and evaluation scripts, is available in the project GitHub repository. A video demonstration walking through the methodology and results is also provided.

GitHub Repository: [SUMMATIVE REPO LINK - GITHUB]

Video Demonstration: [Model Training and Evaluation Video]